

Tokenization, Preprocessing, and POS Tagging

1 Question 2: English Text Preprocessing, Frequency Analysis, and POS Tagging

1.1 Part A: Text Preprocessing

The raw text from Jane Austen's *Emma* was loaded using NLTK's Gutenberg corpus reader and first tokenized as is with NLTK's word tokenizer, with no modification, to obtain a raw baseline token count and vocabulary size.

Before computing the after preprocessing statistics, one additional normalization step was applied to the raw text: every occurrence of – (Project Gutenberg's rendering of an em dash, written with no surrounding whitespace) was replaced with – surrounded by spaces. This step is necessary because without it the tokenizer fuses the final word of a sentence together with the following em dash into a single token instead of treating them as separate tokens.

After this normalization, every token was lower cased and passed through two filters: first an alphabetic check to strip out punctuation, digits, and any other non-word tokens and then a stop word filter using NLTK's built-in English stopword list.

Table 1: Token and vocabulary counts before and after preprocessing

	Total tokens	Unique words
Before preprocessing	191,855	8,376
After preprocessing	70,171	6,831

Preprocessing removed 63.42% of all tokens but only about 18.45% of the unique word types. This gap exists because tokens such as stop words and punctuation marks repeat the most in the novel so removing them decreases the token count a lot while barely affecting the size of the underlying vocabulary.

1.2 Part B: Word Frequency Analysis

This section identifies the ten most frequent words remaining after preprocessing, computed with a simple frequency counter over the filtered token list.

Table 2: Top 10 most frequent words after preprocessing

Rank	Word	Frequency
1	emma	865
2	could	837
3	would	819
4	miss	599
5	must	567
6	harriet	506
7	much	486
8	said	484
9	one	447
10	weston	439

The number of words occurring exactly once, known as hapax legomena, is 2,682, which is around 39.26% of the 6,831 unique words left after preprocessing.

There are two things that stand out the most in this list. First, the character names *emma*, *harriet* and *weston*, along with the address term *miss*, take up four of the ten slots, which is expected in a novel built around characters and their social interactions. Second, the modal verbs *could*, *would* and *must* survived stopword removal and landed in the top five most frequently repeated words. This shows NLTK's default English stopword list does not include modal auxiliaries, so words that function more like grammatical connectors than content words still make it through the filter.

1.3 Part C: Part-of-Speech Tagging

Nouns are defined as any token tagged NN, NNS, NNP or NNPS, i.e. singular and plural common nouns together with singular and plural proper nouns. Proper noun tags are included because a large share of the meaningful nouns in this text are character names and excluding them would have made the noun analysis mostly about generic objects rather than the people the story is actually about.

Before tagging, the em-dash preprocessing step was applied: every occurrence of – was replaced with – surrounded by spaces. Without this step, the tokenizer adds a sentence-final period onto the word immediately before an em dash – for example “...brought it.–Nobody...” becomes the single token “it.” instead of the separate tokens “it”, “.”, “–” and the POS tagger can then mistag that fused token as a noun even though the underlying word is a pronoun, preposition, adverb or verb. Adding the spaces back restores normal token boundaries and avoids this class of error.

POS tagging was performed on this em-dash-normalized text prior to lowercasing or stopword removal rather than on the preprocessed token list. It was performed sentence by sentence rather than on one long list of tokens. Tagging on the filtered, lowercased token list removed exactly the two signals the tagger relies on: capitalization (needed to recognize proper nouns and titles) and sentence level context (needed to resolve ambiguous words) so tagging was deliberately done before those steps.

In addition to the tag based filter, each noun token was required to match the pattern of one or more alphabetic characters with an optional single trailing period. This excludes Project Gutenberg formatting artifacts such as underscore-delimited italics (“_her_”, “_elton_”) and stray bracket punctuation (“[”, “]”) from being counted as nouns while still allowing genuine abbreviations like “Mr.” and single letter initials to pass through. After tagging, every noun was lower cased before being counted so that capitalized and lowercase occurrences of the same word (for example *Time* and *time*) are not counted as two different nouns.

The total number of noun occurrences across the text is 31,317, and the number of distinct nouns is 3,711.

The titles *mr* and *mrs* appear without their trailing period because of an additional normalization step: for every noun token ending in a period, the code checks whether the same word also occurs elsewhere in the noun list without a period. Sentence boundary

Table 3: Top 10 most frequent nouns (NN, NNS, NNP, NNPS)

Rank	Noun	Frequency
1	mr	1,153
2	emma	857
3	mrs	699
4	miss	584
5	harriet	494
6	weston	439
7	thing	398
8	knightley	389
9	elton	385
10	woodhouse	311

detection occasionally splits “Mr.” into “Mr” and “.” as two separate tokens, particularly when the title is immediately followed by a name across a mis-detected sentence break, so the bare form “mr” does genuinely appear in the raw noun list alongside the far more common dotted form “mr.”. When a bare form is found, the two are merged into a single count under the bare spelling; if no bare form exists anywhere in the noun list (for example “Dr.” or “Sept.”) the period is kept, since it is a stable part of the token. This is why the titles in Table 3 report the full, unsplit frequency (1,153 for *mr* and 699 for *mrs*) rather than an undercount split across two spellings of the same word. Aside from these two titles, the list is almost entirely character names (*emma*, *harriet*, *weston*, *elton*, *knightley*, *woodhouse*) plus the address term *miss* and the generic noun *thing* which is common in a dialogue heavy novel.

1.4 Part D: Analysis

Effect of preprocessing on vocabulary and frequency.

Preprocessing reduced the token count by about 63.42% but the unique word count by only about 18.45% (see Table 1). This large gap is expected for stopword removal specifically since a handful of function words like *the*, *to* and *and* account for a huge share of raw tokens but are each only one vocabulary entry so removing them shrinks the token count far more than the word type count.

Effect of stopword removal on the most frequent words.

Stopword removal did substantially change the list since none of the function words that would dominate an unfiltered frequency count such as *the*, *to* and *and* survive it. It is not a complete fix though because NLTK’s default stopword set omits modal auxiliaries like *could*, *would* and *must* which is why they still occupy half of the top five in Table 2 even after filtering. This suggests that the most frequent word and the most meaningful word are not the same thing even after standard stopword removal and shows that the choice of stopword list has a direct effect on the frequency table.

Types of nouns that dominate.

Excluding the two honorifics, the list in Table 3 is almost entirely the same six character names already noted in Part C, plus *miss* and *thing*. Only one of the ten entries, *thing*, is a generic common noun, which fits a novel whose plot is driven almost entirely by conversation and social standing among a fixed circle of characters rather than by description of objects or settings.

Preprocessing decisions that affected POS tagging.

Tagging the raw text before any lowercasing or stopword removal was a deliberate choice: keeping capitalization let the tagger recognize character names and titles such as *Mr.* and *Mrs.* as proper nouns and tagging sentence by sentence rather than on one flat list of tokens preserved the sentence level context the tagger uses to resolve ambiguous words. Lowercasing the identified nouns only after tagging allowed case variants (like *Time* and *time*) to be merged into a single count without affecting the tagging decision itself.

The clearest example of a preprocessing decision affecting tagging is the em-dash handling described in Part C. Fused tokens like “*it.*”, produced whenever a word was immediately followed by – with no space, were occasionally tagged as nouns purely because of this Project Gutenberg formatting quirk and not because the tagger misunderstood grammar; the word *it* on its own is a pronoun, not a noun. Normalizing the em dash before tagging removed this source of error. This shows that tagging errors can come from source text formatting quirks just as easily as from the tagger’s own limitations so checking a frequency list for oddly punctuated tokens is a useful sanity check to run after any tagging pass.